Meehir Prabhakar

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Education

VIT-AP University India

Bachelor of Technology - BTech, Computer Science and Engineering *Aug. 2023 – Expected Jun 2027*

Experience

TechtoGreen Drone & Robotics Private Limited Amaravati, Andhra Pradesh, India

AI/ML and Computer Vision Intern *Dec. 2024 – May 2026*

- **AI Engineering & Solution Development:** Engineered and tested multi-model AI pipelines (YOLO, VGGT, OpenCV) for real-time inference, deploying scalable AI-powered solutions on edge devices.
- **Software Engineering & API Development:** Developed robust backend services using FastAPI to ingest live telemetry and automate cross-platform reporting, seamlessly integrating AI capabilities into business workflows.
- **Data Engineering & Model Enablement:** Built and maintained data pipelines to process structured MAVLink telemetry and unstructured point cloud data (YDLIDAR), ensuring data quality and readiness for model evaluation.
- **Optimization & Tracking:** Designed custom Non-Maximum Suppression (NMS) algorithms, implemented dynamic GPU memory allocation, and tracked model performance to ensure high reliability and continuous improvement.

The Machine Learning CLUB VITAP Amaravati, Andhra Pradesh, India

Technical & Project Team Member *Oct. 2023 – May 2025*

- **Solution Design & Collaboration:** Translated complex requirements into scalable technical solutions, utilizing NavigationMesh for spatial pathfinding and dynamic AI behavior.
- **Development:** Engineered interactive environments with state-persistent UIs using modern frameworks and GDScript, collaborating with cross-functional teams on design reviews.

Projects

AI SIEM & MDR Pipeline | *Python, Scikit-Learn, SHAP, LIME* Jan. 2026 – May 2026

- **Data Processing & Preprocessing:** Analyzed unstructured datasets (network flows, OS logs) and built pipelines to clean and normalize heterogeneous telemetry into a unified JSON schema for ML training.
- **Machine Learning Models:** Trained and evaluated Isolation Forests and One-Class SVMs (with TF-IDF N-grams) for robust anomaly and cyber threat detection.
- **Documentation & Explainability:** Integrated SHAP and LIME to interpret model architectures, track performance, and provide actionable insights for threat correlation.
- Hosted on GitHub: github.com/Meehir5907/Chakshu

Autonomous Sewage Maintenance Submarine | *Python, YOLOv8n, NCNN, Open3D, RPi 5* Jul. 2024 – Nov. 2025

- **Model Deployment:** Deployed NCNN-optimized YOLOv8n segmentation models on headless hardware (Raspberry Pi 5) for real-time structural analysis.
- **Algorithm Design:** Analyzed mechanical SONAR data using Open3D noise reduction and bicubic interpolation to build interactive depth heatmaps.
- Programmed 3D propulsion and arm actuation via `lgpio`, controlling RS2205 brushless motors via 30A ESCs and a PCA9685 PWM controller.

Awards & Achievements

1st Place (Dept. of AI/DS), Innovative Model: Secured 1st place at Siddhartha Academy of Higher Education Innovation Day (Oct. 2025) in department of Artificial Intelligence / Data Science.
AI Hackathon 2025: AIMERS Letter of Appreciation (Feb. 2025).
Ideathon 2K25: Innovation Showcase Participant at Chalapathi Institute of Engineering & Technology (Mar. 2025).

Technical Skills

Core Languages: Python, C, C++, SQL, Bash, Git
AI & ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLO, Transformers, FAISS, SHAP, LIME
Software & Data Engineering: API Development (FastAPI), Data Pipelines, CI/CD Concepts, Linux (KVM, Kernel Tuning)
Core Domains: AI Engineering, Machine Learning, Data Preprocessing, Computer Vision, Model Deployment, Agile Methodologies